8.033 Fall 2025 Solution Set 1

1 Review: curves and tangents in $\mathbb{R}^3$

a.

$$\frac{d\mathbf{v}}{d\lambda} = \left(2\lambda\cos\left(\lambda^2\right), \frac{1}{\lambda+1}, 4\lambda\right).$$

b.

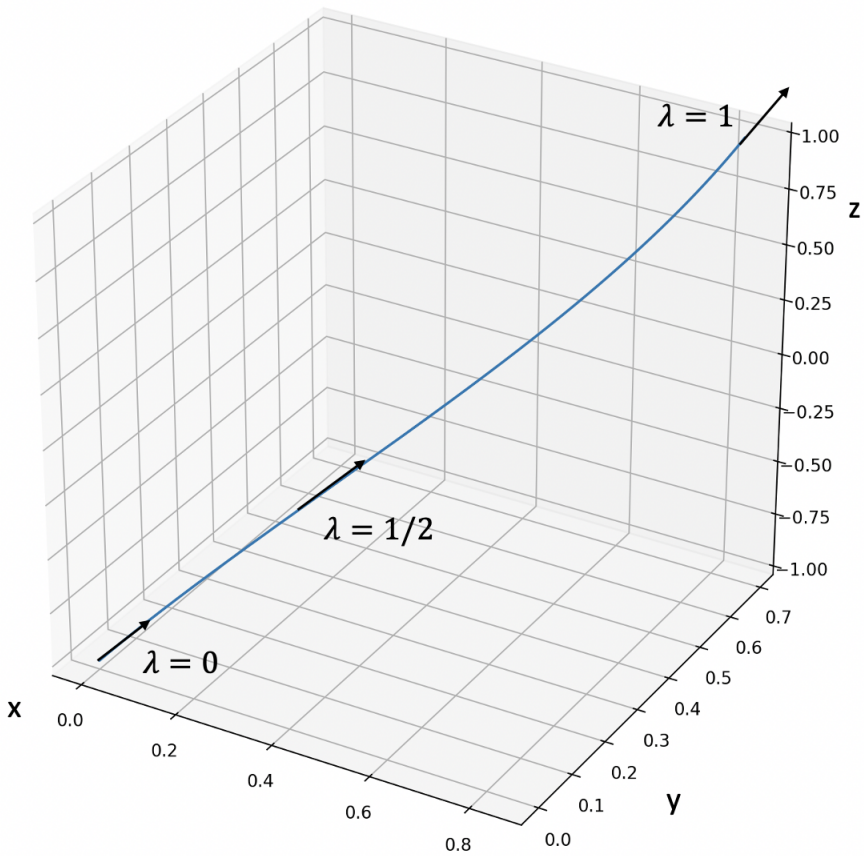

Figure 1: The parametric curve and tangent vectors at $\lambda = 0$, $\lambda = 1/2$, and $\lambda = 1$.

c.

$$r = (v_x^2 + v_y^2 + v_z^2)^{1/2} = [\sin^2\left(\lambda^2\right) + \ln^2\left(\lambda+1\right) + (2\lambda^2-1)^2]^{1/2},$$
$$\theta = \cos^{-1}\left(v_z/r\right) = \cos^{-1}\frac{2\lambda^2-1}{[\sin^2(\lambda^2)+\ln^2(\lambda+1)+(2\lambda^2-1)^2]^{1/2}},$$
$$\phi = \tan^{-1}\left(\frac{v_y}{v_x}\right) = \tan^{-1}\left(\frac{\ln(\lambda+1)}{\sin(\lambda^2)}\right).$$

d.

$$\mathbf{v}\cdot\mathbf{v} = |\mathbf{v}|^2 = \sin^2\left(\lambda^2\right) + \ln^2\left(\lambda+1\right) + (2\lambda^2-1)^2.$$

e.

$$\begin{aligned}\mathbf{v}\times\mathbf{v}' =&(4\lambda\ln(\lambda+1)-(2\lambda^2-1)/(\lambda+1),\\ &(2\lambda^2-1)\cos(\lambda^2)2\lambda-4\lambda\sin(\lambda^2),\\ &\sin(\lambda^2)/(\lambda+1)-\ln(\lambda+1)\cos(\lambda^2)2\lambda).\end{aligned}$$

2 Review: Short Problems

a.

$$\begin{aligned}\frac{\partial f}{\partial x}&=\cos(x(\lambda))-\frac{y(\lambda)}{x^2(\lambda)},\\ \frac{\partial f}{\partial y}&=\frac{1}{x(\lambda)},\\ \frac{\partial f}{\partial \lambda}&=\cos(x(\lambda))\frac{\partial x}{\partial\lambda}-\frac{y(\lambda)}{x^2(\lambda)}\frac{\partial x}{\partial\lambda}+\frac{1}{x(\lambda)}\frac{\partial y}{\partial\lambda}+2\lambda.\end{aligned}$$

b.

$$\frac{1}{(1-x^2)^{1/3}}=1+\frac{1}{3}x^2+\mathcal{O}(x^3).$$

c.

$$f(x)=A\sinh(\sqrt{2}x)+B\cosh(\sqrt{2}x).$$

3 Review: Rotation Matrix

a.

$$\begin{aligned}x_0'&=x_0\cos\theta-y_0\sin\theta,\\ y_0'&=x_0\sin\theta+y_0\cos\theta,\\ z_0'&=z_0.\end{aligned}$$

b. Let

$$R[\theta]=\begin{bmatrix}\cos\theta & -\sin\theta & 0\\ \sin\theta & \cos\theta & 0\\ 0 & 0 & 1\end{bmatrix},$$

then it is easy to check that $X'=R[\theta]\cdot X$, where

$$X'=\begin{bmatrix}x_0'\\ y_0'\\ z_0'\end{bmatrix},$$

and

$$X=\begin{bmatrix}x_0\\ y_0\\ z_0\end{bmatrix}.$$

4 Review: Matrix Multiplication

a. Let $[A] = [M] \cdot [N]$. Then $A_{ij} = \sum_k M_{ik} N_{kj}$.
b. Let $[B] = [N] \cdot [M]$. Then $B_{ij} = \sum_k N_{ik} M_{kj}$.
c. $\mathbf{v} \cdot [M] \cdot \mathbf{u} = \sum_j \sum_i v_j M_{ji} u_i$.
d. $\mathbf{v} \cdot \mathbf{u} = \sum_i v_i u_i$.

5

Suppose an observer A has a device that can apply a constant force $\mathbf{F}_A$ on a test mass m. Since Newton's second law holds in A's frame, we have $\mathbf{F}_A = m\mathbf{a}_A$, where $\mathbf{a}_A$ is the acceleration of the test mass observed by A. Now to check that Newton's second law is consistent with the Principle of Relativity, let's consider an observer B who is moving along the x-axis (choosing the x-axis such that $\hat{\mathbf{x}} = \hat{\mathbf{a}}_A$) with a velocity $+v$ relative to A. Since the force exerted by the device on the test mass is independent of the reference frame, we just need to check that $\mathbf{F}_A$ is still equal to $m\mathbf{a}_B$, where $\mathbf{a}_B$ is the acceleration of the test mass observed by B. Using Galilean relativity that relates $x_B = x_A - vt_A$ and $t_B = t_A$, we find that

$$a_B = \frac{d^2 x_B}{dt_B^2} = \frac{d^2(x_A - vt_A)}{dt_A^2} = \frac{d}{dt_A}\left(\frac{dx_A}{dt_A} - v\right) = \frac{d^2 x_A}{dt_A^2} = a_A. \tag{5.1}$$

Since $\mathbf{a}_B = \mathbf{a}_A$, we find

$$\mathbf{F}_A = m\mathbf{a}_B. \tag{5.2}$$

In order words, Newton's second law still holds in B's frame.

To check Newton's law of gravitation, suppose we have two test masses m_1 and m_2 located at $\mathbf{x}_{1,A}$ and $\mathbf{x}_{2,A}$, respectively, in A's frame. Align the x-axis along the direction of $\mathbf{r}_A \equiv \mathbf{x}_{1,A} - \mathbf{x}_{2,A}$. The gravitational force between the two masses observed in A's frame is

$$\begin{aligned} \mathbf{F}_A &= -G\frac{m_1 m_2}{r_A^2}\hat{\mathbf{r}}_A \\ &= -G\frac{m_1 m_2}{(x_{1,A} - x_{2,A})^3}(\mathbf{x}_{1,A} - \mathbf{x}_{2,A}). \end{aligned} \tag{5.3}$$

Now observer B observes in their frame

$$\mathbf{F}_B = -G\frac{m_1 m_2}{(x_{1,B} - x_{2,B})^3}(\mathbf{x}_{1,B} - \mathbf{x}_{2,B}). \tag{5.4}$$

Using Galilean Relativity, we find that

$$x_{1,B} - x_{2,B} = (x_{1,A} - vt_A) - (x_{2,A} - vt_A) = x_{1,A} - x_{2,A}. \tag{5.5}$$

Plugging back to Eq. (5.4), we find that

$$\mathbf{F}_B = -G\frac{m_1 m_2}{(x_{1,A} - x_{2,A})^3}(\mathbf{x}_{1,A} - \mathbf{x}_{2,A}) = \mathbf{F}_A, \tag{5.6}$$

so Newton's law of gravitation is consistent with the Principle of Relativity as well under Galilean Relativity.

6

It makes sense to talk about observer 1 being "at rest relative to" observer 2 or "moving at some velocity" relative to observer 2 because there is no absolute way of determining if an observer is at rest or not. Therefore, it only makes sense to say if one observer is moving relative to another. On the other hand, an observer can tell if her experience is consistent with Newton's law of inertia. This is an absolute definition that makes no reference to any other frame, so it does not make sense to say observer 1 being "inertial relative to" observer 2.

7

a. The parametric equations for the two-dimensional cone in $\mathbb{R}^3$ is

$$\begin{aligned} x &= \frac{r}{h} z \cos\theta \\ y &= \frac{r}{h} z \sin\theta \\ z &= z, \end{aligned} \tag{7.1}$$

where we have $0 \leq z \leq h$ and $0 \leq \theta < 2\pi$.

b. Using the opening angle of the cone, α, (and recall that the opening angle is defined as $\alpha = 2\tan^{-1}(r/h)$) we can rewrite Eq. (7.1) as

$$\begin{aligned} x &= z \tan\frac{\alpha}{2} \cos\theta \\ y &= z \tan\frac{\alpha}{2} \sin\theta \\ z &= z. \end{aligned} \tag{7.2}$$

To specify an infinitely-extended cone, we just need to extend the range of z from previously $[0, h]$ to now $[0, \infty)$.

c. To write down the equations for the three-dimensional cone, we just need to allow the opening angle from Eq. (7.2) to vary between 0 and the cone's boundary, namely

$$\begin{aligned} x &= z \tan\frac{\alpha'}{2} \cos\theta \\ y &= z \tan\frac{\alpha'}{2} \sin\theta \\ z &= z, \end{aligned} \tag{7.3}$$

where $\alpha' \in [0, \alpha]$. It is straightforward to use Eq. (7.3) to prove the statement that we are asked to prove. We pick an arbitrary point inside the cone using Eq. (7.3), namely fixing a value for z, θ, and α', respectively, and call it p. Then if we allow z to vary while fixing θ and α', we get the equation for the line that connects the origin and point p. Since we picked p to be inside the cone, namely requiring $\alpha' < \alpha$, the angle that this line makes with the z-axis (namely $\alpha'/2$) is smaller than $\alpha/2$. Conversely, we can pick a line with $\alpha'/2 < \alpha/2$ by fixing θ while allowing z to vary. Then any point on this line has $\alpha' < \alpha$, and by Eq. (7.3) we know such a point is located inside the cone, so we are done with the proof.

8

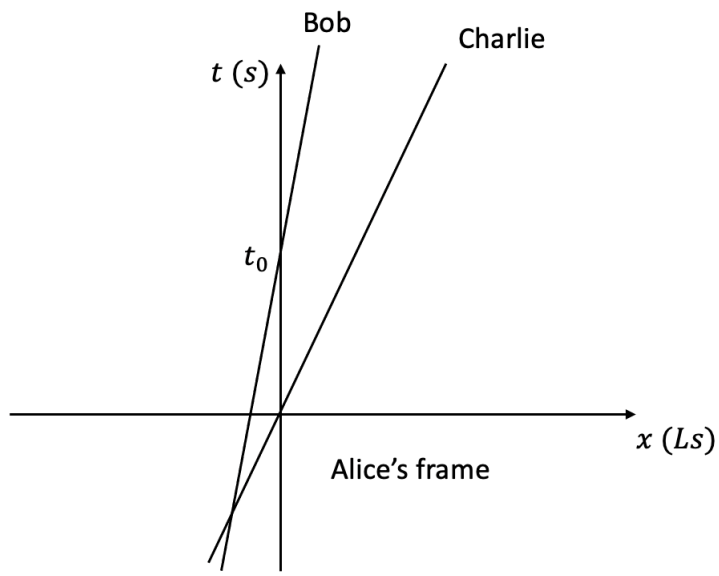

Figure 2: Bob and Charlie's worldlines in Alice's frame. Note that since $v_{CA} > v_{BA}$, the slope of Bob's worldline is steeper than Charlie's and both > 1.

Assume that Bob's clock reads 0 when Bob is at $(t_A = t_0, x_A = 0)$ reviewed in Alice's frame. Then by Galilean Relativity, Bob's and Alice's clocks are related by

$$t_B = t_A - t_0, \tag{8.1}$$

since when Alice reads time t_0, Bob reads time 0. Then the relationship between the position of an event viewed by Alice, x_A, and the position of an event viewed by Bob, x_B, is

$$x_B = x_A - v_{BA}(t_A - t_0). \tag{8.2}$$

We have $(t_A - t_0)$ multiplied with the velocity because when Alice reads time t_0, Bob's spacial coordinates coincide with Alice's.

Relating the coordinates seen by Charlie and those seen by Alice is straightforward:

$$\begin{aligned} t_C &= t_A \\ x_C &= x_A - v_{CA} t_A. \end{aligned} \tag{8.3}$$

Rearrange Eq. (8.2) and plug into the second line of Eq. (8.3), we get

$$x_C = x_B + v_{BA}(t_A - t_0) - v_{CA} t_A = x_B - v_{BA} t_0 + (v_{BA} - v_{CA}) t_C, \tag{8.4}$$

where in the last equality we have used $t_C = t_A$. From Eq. (8.4), we can find the relative velocity between Bob and Charlie by reading off the term appearing with t_C, which is $v_{BA} - v_{CA}$. It is indeed independent of the time off-set t_0.

9

Suppose B lies in the future of A, then by the statement given at the beginning of the problem, B lies in the solid cone bounded by light rays emitted from A. The conclusion of problem 7(c) states that any point p inside a solid cone lies on a line from the origin to p, where the angle that this line makes with the z-axis is smaller than half of the opening angle of the cone. This means B lies on a line from A to B, and the angle this line makes with the z-axis is smaller than 45 degrees. Namely, there is a single (sublight) worldline that passes through A and then through B.

Now we prove the other direction. Suppose there is a single sublight (i.e. the angle with the z-axis is smaller than 45 degrees) worldline that passes through A and then through B, then the conclusion of problem 7(c) indicates that B, as a point that lies on such a line, must live inside of the solid cone. Then by the statement given at the beginning of this problem, B in the solid cone bounded by light rays from A indicates that B lies in the future of event A, so we are done.

10

a.

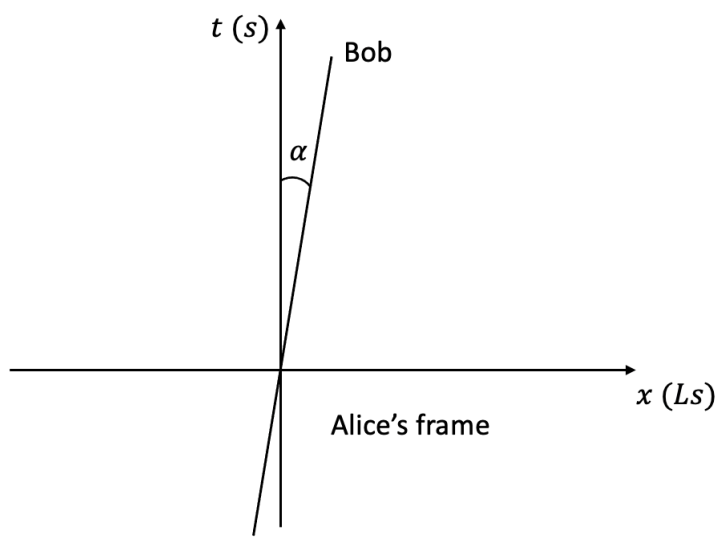

Figure 3: Bob's worldline in Alice's frame, making an angle α with the time axis.

See Fig. (3). The angle that Bob's worldline makes with Alice's t-axis is given by

$$\tan\alpha = \frac{x}{ct} = \frac{0.5ct}{ct} = 0.5, \tag{10.1}$$

so $\alpha = \tan^{-1} 0.5$.

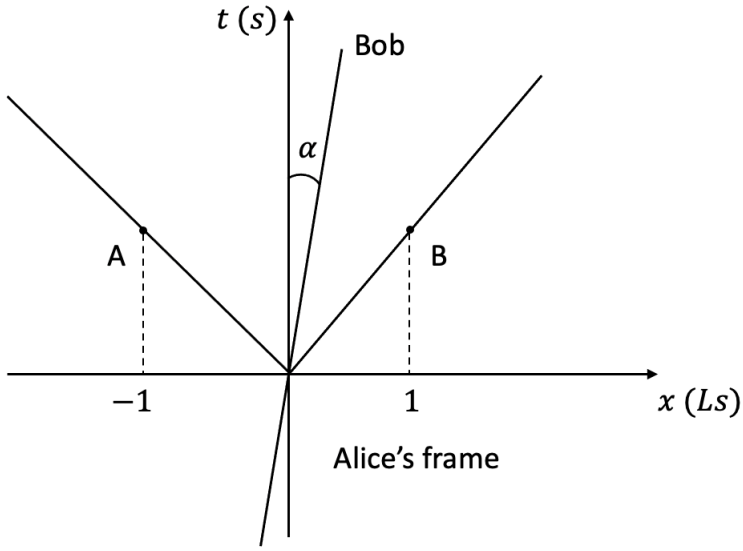

Figure 4: A flashlight is turned on at the origin. Events A and B are one second after the burst of the light on the the left and right light rays, respectively, seen in Alice's frame.

b. See Fig. (4).
c. See Fig. (5).

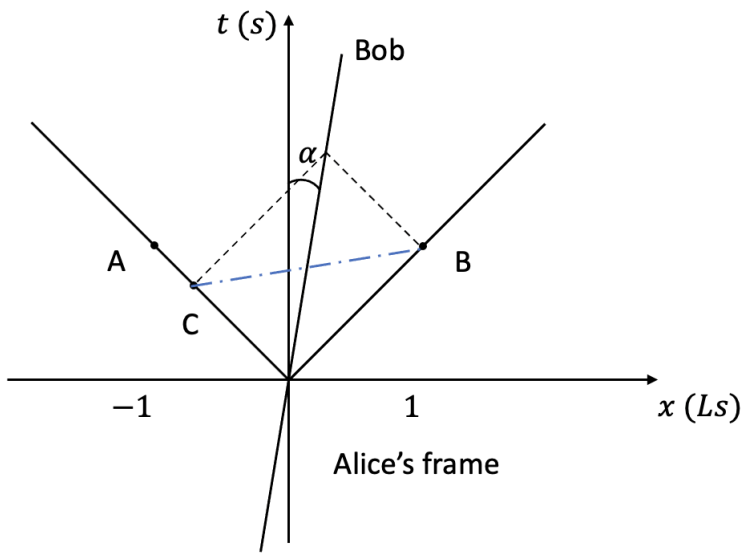

Figure 5: We find event C, the event on the left light ray that is simultaneous with B as determined by Bob, using the definition of simultaneity. Namely, events B and C are simultaneous in Bob's frame if light rays from Bob's worldline (also the t-axis of Bob's frame) at a given time bounce off events C and B and together return to Bob's worldline at the same event. The dashed lines are light rays bouncing back from events C and B, and the dash-dotted line is Bob's line of simultaneity that connects events C and B.